

JHABUA POWER LIMITED (A JV of NTPC Ltd)

Village- Barela, Tehsil- Ghansore, District- Seoni

State- Madhya Pradesh ,

Pin Code- 480997 (M.P.) INDIA

Implementation and Support

of

SAP S/4HANA ERP System

at

JPL NTPC

Technical Specifications and Scope of Work

(THIS DOCUMENT IS MEANT FOR THE EXCLUSIVE PURPOSE OF BIDDING AGAINST THIS SPECIFICATION AND SHALL NOT BE TRANSFERRED, REPRODUCED OR OTHERWISE USED FOR PURPOSES OTHER THAN FOR WHICH IT IS SPECIFICALLY ISSUED)

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Clause No	Description
1.0	INTRODUCTION
1.1	General Overview Jhabua Power Limited (JPL) is a JV of NTPC Limited, operating as JV since 5th September 2022, JPL running with 600 MW unit and expansion of 660 MW unit. Need is being felt to migrate from the existing SAP ECC to the latest version of SAP S4 HANA “RISE” latest version with additional functional modules. JPL is Planning for green field implementation of S4 HANA - covering the existing running SAP modules and additional functional modules. The existing ECC system will be treated as legacy system and the open items will be moved to S4 HANA while cutover activities. JPL NTPC had implemented ERP on SAP ECC6 at its Primary Data Center (PDC) at SIFY DC Noida and Disaster recovery center (DRC) at SIFY DR Aroli Mumbai. The SAP-ERP environment of JPL NTPC is hosted on Unix Servers running on IBM-AIX with Oracle databases. Existing SAP ERP solution was implemented in year 2011, covering various functionalities such as FI, MM, SD, HCM, Payroll, PS and Employee Self Service (ESS) JPL NTPC has a three-tier system landscape consisting of Dev, QAS, and Production systems. . As JPL NTPC intends to move from ECC to SAP S/4HANA system, the complete S4 HANA landscape needs to be created on RISE covering the existing modules as well as additional modules as mentioned

	later in this document. For this, JPL NTPC invites bids for “Implementation and Support of SAP S/4HANA ERP System” as per the technical requirement and Scope of Work detailed in this document. The bids shall be prepared and submitted in accordance with the provisions in the bidding document. Noncompliance may result in rejection of bid. This document contains Scope of Work, Technical Specifications, Bidder’s Data requirement sheets, Conditions of Contract and Bill of Quantity. This document is meant for the exclusive purpose of bidding against this specification and shall not be transferred, reproduced or otherwise used for purposes other than for which it is specifically issued.
1.2	Limit of Contract The bidder should ensure that the solution offered is complete in every respect and adhere to the project plan which will be prepared during the project preparation phase and approved by JPL NTPC. It shall be the responsibility of the bidder to include all hardware, software components, modules, tools and services, whether specified or otherwise, that would be needed to meet the intent and requirements of providing a total solution to JPL NTPC within the quoted price.
1.3	Information to be furnished by the bidder The Bidder must furnish complete information asked for in the bidding document, complying explicitly with all the clauses/requirements in the bidding document. Incomplete information/bid may lead to rejection of the bid. The Bidder must furnish the requirements/ specifications regarding the following, amongst others, along with its bid in the specified order: i) A profile of the company. ii) Details regarding the manpower, tools, and infrastructure available with the bidder responsible for carrying out implementation and maintenance at site. iii) Bidder has to submit an authorization from M/s. SAP India to participate in the tender. iv) Specifically for Contract 2 - Bidders are required to quote for <i>all</i> the items of the Tender Enquiry specified in the technical specification document and incomplete offers will be rejected. The quoted price for five-year support should not be less than 20% of the total bid price. If it is less than 20% then the difference between actual amount and 20% of bid price shall be withheld and released proportionately at the end of each year of support. v) Specifically for Contract 3 : The quoted price for Supply of S4 HANA with RISE License during Implementation and Post GO-Live Support Period including DR (1-Year) and Supply of S4 HANA with RISE License for Four Years after GO-Live including DR (4-Year) combined should not be less than 40% of the total bid price. If it is less than 40% then the difference between actual amount and 40% of bid price shall be withheld and released proportionately at the end of each year. vi) Relevant portions of the documents submitted in pursuance of eligibility criterion shall be highlighted. If the offer is not accompanied by all the documents, the same would be rejected. vii) The bidder must have registered office in India. Documentary proof shall be submitted. viii) Any other document/information as per the requirement of the tender. JPL NTPC reserves the right to verify/ confirm all original documentary evidence submitted by vendors in support of the eligibility criteria. Upon verification, evaluation/ assessment, if any information furnished by the bidder is found to be false/ incorrect, their total bid will be rejected and correspondence on the same shall not be entertained.

2.0	SCOPE OF WORK
2.1	Broad Objective JPL NTPC is currently using SAP ERP System to carry out its day-to-day business processes. JPL NTPC plans to implement SAP's latest generation ERP platform, S/4HANA and transfer required data from legacy system to the new system. This implementation aims to bring benefits such as improved speed, enhanced quality of operations, and leveraging intelligent automation capabilities offered by S/4HANA. The broad objectives of this proposed engagement are: i) To supply S4HANA licence with SAP RISE infrastructure to cater to complete requirement for implementation of all the modules and successfully run the system during implementation and upto 5 years after post go live support period ends. ii) To successfully implement SAP S/4 HANA Solution on RISE with SAP including the functionalities mentioned in the scope. iii) To transfer data from JPL NTPC's current SAP ECC landscape data to new system to maintain continuity as well as speed up implementation. iv) All historical business and transactional data from the existing SAP ECC system shall be archived securely in compliance with applicable legal, regulatory, and organizational requirements. v) The archived data shall remain accessible for a minimum period of five (5) years post-migration, with appropriate tools and mechanisms to retrieve, view, and report on archived information as needed by JPL NTPC. vi) Implementation of new solutions as per the scope of work and FRS. vii) To implement necessary network and security architecture. viii) To leverage capabilities of S/4 HANA such as improved speed & quality of operations, new processes, analytics, FIORI UI, etc. ix) To provide post Go-Live support for stabilization of implemented solutions for a period of Three (3) months. x) To provide post-implementation five-year support.
2.2	Detailed Scope
2.2.1	Organizational Scope This Scope document covers technical upgrade , implementation and migration of JPL NTPC SAP implementation in scope for legal entity as mentioned below: • Jhabua Power Limited (A JV of NTPC Limited).
2.2.2	Project Location • The Project will be implemented and supported from JPL NTPC Plant at District- Seoni, Madhya Pradesh, Pin- 480997 and Noida Office Location JPL NTPC EOC Sector 24 Noida Office • SI's Project Manager and consultants must be available at above locations as per requirement for the entire project duration and its post Go-Live stabilization period. The resource deployment plan must be detailed by bidder during the project preparation phase.
2.2.3	Language and Currencies i) The SAP solution will display all business data in English. ii) Project communication and documentation will be in English. iii) System currency will be Indian Rupee (INR) iv) All business data in the system will be in English language only.

2.2.4

Current System Landscape

The current environment has Application servers and DB servers as per details given below in Table-A. Each server has a purpose and function and has multiple instances like *Development, testing, production* etc., depending on business requirement.

Table-A represents the current SAP landscape of JPL NTPC along with their application version, operation system, database version and availability of respective environments.

Existing SAP Sizing Information landscape sheet enclosed with annexure 1

Table-A

System s	Operating System	DB size	Database	SBX	DEV	QAS	PRD	DR
ECC 6.0 EHP5	IBM Aix Version 7.1	1.5 TB	Oracle 11.2.0.2.0	Y	Y	Y	Y	Y
EP Netweaver 7.3	IBM Aix Version 7.1	68-0GB	Oracle 11.2.0.2.0	Y	Y	Y	Y	Y

The OS and DB versions mentioned in the table may get upgraded as per patch update policy of JPL NTPC during the tendering process. The size of DB may also change.

The ECC and Netweaver as above are currently out of support since 2016-17. Bidder shall be required to archive complete transaction data and store on the new system over cloud for viewing purpose for a period of five years after cut-over and go live. Cost of any software, tools, hardware and licensing if required for this activity shall be in the scope of bidder.

“ SAP License for reinstatement only of exiting SAP ECC system shall be provided by JPL NTPC with coordination of SAP ”. It will not have any relation with the SAP licenses and RISE infrastructure to be supplied by bidder for implementation of new system.

2.2.5

Modules & Solutions to be implemented

The current implementations include but not limited to the following:

A. The modules/solutions in ECC EHP5

a) The existing modules/solutions in ECC6 EHP5

i)

Material Management (MM)
Inventory Management (IM)

ii)

Finance & Costing (FI/CO)

iii)

Project System (PS)

iv)

Quality Management (QM)

v)

Human Capital Management (HCM)
Organizational Management
Personal Administration
Time Management
Travel Management

	vi) Payroll vii) ESS (Employee Self Services) b) The Intended (new functionalities and improvement in the existing functionalities) modules/solutions to be implemented on S4 HANA i) Finance and Costing ii) Energy Billing and Accounting iii) Operations iv) Project and Plant Maintenance v) Project Systems vi) Quality Management vii) Sales Distribution viii) Fuel Management ix) Material Management x) Document Management System xi) HR and Payroll Management B. SAP Integration Suite C. SAP BTP D. ITSM – SAP/ 3rd Party, Cloud ALM (technical monitoring + CHARM)															
2.2.6	Major third-party integrations Table-B given below includes the integrations of current SAP ERP installation with external public portals and internal applications. Table-B <table><tr><th>SI No</th><th>Application</th><th>Functionality</th></tr><tr><td>1.</td><td>FTP Services</td><td>Data export to FTP servers from SAP systems</td></tr><tr><td>2.</td><td>Material Gate pass System</td><td>Material Gate pass issue & return process</td></tr><tr><td>3.</td><td>GEPNIC Integration</td><td>Integration of MM Module with GEPNIC system of NIC (Functional requirement shall be provided by JPL NTPC and SI to Implement the integration as per requirement)</td></tr><tr><td>4</td><td>FOIS System of Indian Railway</td><td>Integration of FIOS system of Indian Railway with JPL NTPC and SI to Implement the integration as per requirement)</td></tr></table> Note: All existing Applications (Serial No.- 1&2), processes and third-party integrations are to be implemented on S/4 HANA even if not mentioned explicitly in the above sections.	SI No	Application	Functionality	1.	FTP Services	Data export to FTP servers from SAP systems	2.	Material Gate pass System	Material Gate pass issue & return process	3.	GEPNIC Integration	Integration of MM Module with GEPNIC system of NIC (Functional requirement shall be provided by JPL NTPC and SI to Implement the integration as per requirement)	4	FOIS System of Indian Railway	Integration of FIOS system of Indian Railway with JPL NTPC and SI to Implement the integration as per requirement)
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2.3	SAP S/4HANA Supply															
2.3.1	Supply of SAP S4HANA License and RISE Infrastructure															

	Bidder will assess S4 HANA license and RISE Infrastructure requirement based on functionality/Modules to be implemented as detailed in this document as well as 140 number of FUE Users except during development phase as detailed in annexure-5 Bidder will be responsible for successful operation of the system for during the full tenure of the contract and shall propose/supply license and cloud infrastructure accordingly. 1. Supply of SAP S/4 HANA Software under RISE as a subscription on Private cloud edition of SAP (including cloud infrastructure, support, and technical services) during implementation and post GO Live Period and subsequently for a period of 5 years. On completion of the SAP S/4HANA RISE subscription, the same can be extended for another two years as per mutual agreement between both the parties. 2. The Bidder shall provide SAP S/4 HANA Private cloud edition to Successfully run all the business applications which requires to be implemented as detailed in this specification having sufficient configuration and resources to efficiently run all the modules and cater to all the users as mentioned in theses specifications. Bidder shall also arrange additional hardware/software resources and tools if required during implementation and subsequent upgradations, sand box for testing etc during tenure of the project, . Offered system should have Development, Quality, and Production (DEV, QAS, PRD) distinct server environments in a system landscape. Two system landscape may be accepted during initial development phases but during deploy and Go-live and afterwards complete three system landscape of adequate configuration shall be provided. 3. Minimum bill of material for SAP S4HANA on RISE licence to be provided is attached at Annexure -5 4. The bidder shall provide all the services required for SAP Basis activities include installing, configuring, and upgrading SAP systems, managing user access and authorizations, monitoring system performance and availability, troubleshooting technical errors, backing up and restoring data, and maintaining the underlying operating system, database, and network infrastructure to ensure the overall health and stability of the SAP environment during complete tenure of the contract.
2.3.2	Scope for SAP S/4 HANA Private Cloud Edition on subscription a. General Scope 1. Provisioning Private cloud environment of SAP S/4 HANA ERP solution to JPL NTPC. All the Data centers which are used for hosting under this contract should be SAP Cloud 2. Provide all the necessary software including SAP S/4 HANA software and necessary licenses on subscription basis comprising of software, software support, cloud Infrastructure, system management services, Technical Manage Services, etc. 3. All the necessary ERP licenses for SAP S/4 HANA software, OS, Network, Antivirus, backup etc. must be provided by bidder. 4. Providing 24x7 technical monitoring and operations support for Production system including DR of SAP Private Cloud. 5. Bidder must fulfil technical delivery and will be responsible for system monitoring and maintenance (alerts, technical upgrades, installations, etc.). 6. Bidder must provide network security for RISE environment. 7. ERP data must be secured by proper encryption and decryption mechanism. 8. Basic operating system installation, Installation of operating System (OS) patches and security updates, Installation of antivirus software and patterns updates, Antivirus software licenses, plan and conduct

	managed service, monitor disk & server capacity, manage data files, file systems Storage and disks, etc. are under bidder's scope. 9. Monitoring of hardware on hypervisor level, monitoring of operating system of provided OS instances, Hypervisor licenses etc. shall be done by the bidder. 10. Offered solution must have Embedded Analytics capability. 11. Provisioning of Data center on Private cloud environment with DR (Disaster Recovery) site at different geographic location and maintaining the same. Bidder shall establish DR set up from Go-Live date with RTO of 4 Hours and RPO of 30 Mins Bidder shall Maintain adequate data security measures as per guidelines of SAP prescribed security, consistent with industry standards and technology best practices, to protect JPL NTPC ERP Data from unauthorized disclosure or acquisition by an unauthorized person. 12. Bidder shall maintain data confidentiality and not to share/disclose any of the information stored on the SAP Private cloud with any person, firm, or organization. 13. JPL NTPC data should not be used for personal or commercial use by any means. 14. The SAP Private cloud services must provide high availability and load balancing capability inbuilt within the SAP Private Cloud Platform and it should be in India. 15. The Bidder shall arrange backup of the data for safe keeping and shall provide full valid backup of ERP data to JPL NTPC for future use after completion of Tenure of this contract. 16. In case of any dispute between current SAP Private Cloud and M/s. SAP affecting the current contract, Bidder shall be responsible for migration of entire system to other CSP who is conforming to the condition of point 1.a.1, without any cost to JPL NTPC. In case of existing CSP is de-empanelled from Meity, Bidder with help of M/s. SAP shall be responsible to migrate the entire system to new CSP who is conforming to the condition of point a.1 of 2.3.2, as per mutually agreed terms and conditions. 17. The successful Bidder should ensure daily incremental and weekly full backup taken for all instances at DC site. The backup should be tested periodically. All backup should be retained as per JPL NTPC backup policy (as enclosed). Any deviation from the backup policy of JPL NTPC as provided shall be duly approved by owner before implementation. In case, DR site activated for any reason, the backup & restoration operation need to be done on DR site till DC site is made ready. 18. On completion of tenure of contract, once the data is backed up and provided to JPL NTPC, given confirmation of the same by JPL NTPC, related data on cloud must be removed/deleted from all the sources. 19. The provided ERP solution should be accessible to JPL NTPC from the cloud data center using VPN/MPLS/Lease Line/Internet. 20. The Bidder must examine the total scope of work of this project and supply the appropriate licenses and required Cloud Infrastructure to run all the business applications which requires to be implemented as detailed in this specification. The number of FUE users should be taken as 140 21. Data Migration server & storage as required should be provided for a minimum period of 12 months or Go-Live date whichever is later. 22. For any performance issue in any of the system in SAP S/4 HANA Landscape, Bidder shall carry out root cause analysis and rectify the same whether the issue is in the landscape or the issue is due to application malfunctioning and do the needful to resolve the issue. 23. Existing SAP landscape shall be migrated to cloud and retained as a reference client which shall not be used for any transaction activity and shall be use for viewing purpose for next 5 Years (From Go-Live) .
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	24. The bidder shall provide security setup as detailed in clause 33 below 25. In addition to standard backup/restore feature offering by M/s. SAP (as per Annexure-4), the following backups are also covered in the scope of work a. Weekly backup of Dev. and Production systems with retention period of 1 month . b. Monthly backup of Dev. and Production systems with retention period of 1 year or end of contract period whichever is earlier. C. Yearly backup of Production systems to be retained for a period of 4 years or till the end of contract period. 26. In case of exigency requirement or drill activity for restoration, on intimation from JPL NTPC for restoration of backup data, the same shall be done by bidder by creating, configuring a Sandbox system in RISE environment and restoring in the system. Maximum number of such restorations will be once in a year. Additional restorations if required shall be done on mutually agreed cost basis. 27. The bidder should maintain monthly availability of solution as below. a. 99.7% for Production Systems. b. 95% for Non-Production Systems. 28. Service Delivery: 24x7 for complete landscape. Service level and penalty as per SLA document of M/s. SAP. 29. The functional scope of offered S4 HANA License and RISE cloud infrastructure should cover all the modules as mentioned in section 2.2.5 and functional requirement FRS annexure 2 30. Additional Application/Solutions to be included in scope of solutions are as follows: a. Cloud Platform Integration (CPI) b. BTP 31. The ERP solution should provide for integration with: a. In-house developed applications which are currently deployed b. Commercial third-party applications which are currently deployed c. Future Integration with In-House /3rd party applications. 32. Domain and DNS Security 1. Domain use and security a. Subdomain of existing parent domain (jhabuapower.co.in) . shall be used for DNS integration with SAP landscape. b. Single parent domain (jhabuapower.co.in)) shall allow integration of the Internet facing SAP portals through Cloud based WAAP and Bot mitigation solution and monitoring of any phishing/squatting attempts and response actions such as take downs of domains, reporting etc. The integration shall be done by NTPC Security team. c. <u>Required integration support (limited to configuration at SAP end if any) for the above shall be provided by SAP.</u> 2. DNS architecture and security a. External DNS resolution shall be done by NTPC Cloud based anycast authoritative DNS servers providing improved access times. The DNS entries shall be suitably configured to allow integration with CWAAP referred earlier to provide adequate Application and DDoS mitigation. The integration shall be done by NTPC Security team. b. Internal DNS resolution: Single On-premises (in HA) DNS server shall resolve entries of SAP servers and shall be maintained by NTPC.
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	The DNS server hosted in SAP tenant shall be in sync with the above On-prem DNS server. <u>Required integration support shall be provided by SAP</u> 33. Security Setup requirement in RISE with SAP A. IDAM and SSO Requirement: 1. Browser based and Fiori'sed SAP applications shall be accessed by integration with Azure AD (Service already owned by JPL NTPC). SSO integration shall be done with Azure AD and Microsoft portal already in use by JPL NTPC. 2. In line with the present practice, Access through SAP GUI is required to be protected and integrated with SSO in view of the proposed use of SAP GUI in future also. "SAP Secure login service for SAP GUI" shall be required for enabling MFA & X.509 certificates-based authentication and SSO integration. 3. <u>All components hosted in SAP RISE environment shall be integrated with JPL NTPC IDAM solution (Present solution is Azure AD and SSO).</u> In case any additional licenses are required to achieve the above functionality, and is not specifically mentioned above, same shall be considered in scope of the bidder and required licenses/services/subscriptions/components shall be quoted by the bidder and provided to NTPC. 34. Security Governance and audit requirements for the proposed solution supplied by the bidder shall be as per annexure 1
2.4	SAP S/4HANA Implementation
2.4.1	Systems Implementation scope The scope shall consist of core systems and auxiliary systems to be implemented by the System Integrator (SI) to S/4HANA RISE with SAP as given in Section 2.2.5. Bidder shall provide SAP licenses and hosting systems for S/4HANA on cloud (RISE with SAP) and the System Integrator (SI) shall undertake the service for migration/implementation of all the functionalities envisaged to achieve intent of the tender. SI shall gather any other information from JPL NTPC ERP core team members, if required, to understand the JPL NTPC's requirement. SI shall provide any additional SAP or third-party related tools and/or licenses pertaining to these Implementation/upgrade activities, at its own cost. The bidder shall carryout implementation of relevant functionalities to new S/4HANA framework as per the information provided in in this document and requirement gathered from JPL core team during preparation and explore phase. • S/4HANA Implementation include but not limited to the following: • Implementation of SAP S/4HANA on RISE in Private cloud • Implementation of all the functionalities as detailed in attached FRS at Annexure - 2 • Implementation of SAP DMS on RISE • Implementation/Migration of all functionalities of EP/ESS to FIORI • Implementation of Payroll • Implementation of all required HR process which are not natively available in bidder supplied HCM module on BTP platform • Custom Code Analysis and Adjustment for compatibility • Implementation of S4HANA Integration Suite

	• Implementation on FIORI of ESS and Travel Module • Master Data Governance Bidder shall fill the relevant columns of FRS and submit along with the bid indicating strategy of implementation for each of the functionalities mentioned. <i>Any work, even if not specifically mentioned but reasonably implied for the system's successful implementation and good performance, is deemed to be included in the scope of the SI.</i>
2.4.2	Existing Source Servers and Proposed Destination Servers for Data Migration The servers present in the current SAP landscape are hosted in DC, which is located at Noida and 100% DR is located at Airoli Mumbai. SI shall provide SAP S/4HANA cloud infrastructure (SAP RISE Model) over SAP Private Cloud for proposed destination servers for Data migration.
2.5	Scope of Implementation/Conversion/Migration, PGLS (Post Go-Live support)
2.5.1	Phases of Implementation/Conversion/Migration The SI shall follow the migration approach and methodology as detailed below to ensure the effective, smooth, and timely completion of the project. The migration processes shall be carried out very systematically and in a streamlined fashion to avoid unexpected delays, technical errors and to minimize downtime. The SI shall ensure support of OEM (SAP) for finding solution to any critical/unresolved issues. The implementation process shall have the following phases: i) Project Preparation Phase ii) Explore Phase iii) Realize Phase iv) Deploy Phase. v) Go-Live Phase vi) Stabilization Phase
2.5.1.1	Project Preparation Phase A. The Project Preparation phase will provide initial planning and preparation for a project. SI will prepare a detailed Project Charter consisting of Project objectives, scope, deliverables, project plan including milestones and time-schedules (L2/L3 Schedule), quality control plan, change management strategy, roles & responsibilities, project structure & resource deployment plan, requirements from JPL NTPC, project risk & mitigation plan, communication matrix, Issue escalation matrix, project templates and project KPIs etc. to measure project success. The Project Charter should also clearly specify the constraints and benefits of the project. B. The Project Charter shall be submitted to JPL NTPC for approval. Macro level and micro level project plan shall be prepared by the SI for all the activities listed in the project. C. The SI will follow SAP recommended project management methodology. SI is required to follow the approved project plan. This plan will serve as the basis for tracking progress during the implementation phase. If the project falls behind schedule, the SI will be obligated to furnish a catch-up plan to address the delay. SI

	shall prepare Project Governance Structure in consultation with JPL NTPC as per details mentioned in sec 2.4.3 and submit for approval of JPL NTPC. D. Resource Planning & deployment: The SI will have to deploy their resources as per their detailed project structure & resource deployment plan. Since the continuity of the key members of the project team is essential for the project's success, NTPC expects SI to follow a diligent process for ensuring this. SI will not be allowed to withdraw/remove any personnel from the project without explicit written confirmation from JPL NTPC and the person withdrawn shall be replaced with an equally or more qualified and competent substitute after approval from JPL NTPC. Complete knowledge transfer about this specific project has to be ensured by SI before the person is replaced. More information on Resource planning & deployment is provided in Sec 2.5.7 & 2.5.8. E. Project Kick-off meeting: SI will conduct the kick-off meeting with JPL NTPC team to explain the Project Charter and scope among other aspects. SI shall also introduce the project team members. F. Overview Training: SI shall provide product overview training to JPL NTPC key stakeholders and key users to familiarize them with product modules, functionalities, nomenclature etc. G. The training calendar for imparting training to Core teams, Power users, end users and top management shall be mutually finalized by both Project Managers during the Project Preparation Phase. H. SI has to submit an Exit Management Plan.
2.5.1.2	Explore Phase A. Understand the AS-IS of JPL NTPC system landscape, business processes through available documents and discussion with JPL NTPC ERP Core Team. B. Finalize Implementation strategy for all the proposed module in the scope . C. Finalize strategy for EP/ESS functionalities implementation on FIORI UI. D. Finalize strategy for developments/enhancements, new process implementations, FIORI adoption etc., in consultation with JPL NTPC. E. Identify existing business process issues, areas of improvement and opportunities for standardization. F. Design the TO-BE processes based on JPL NTPC business requirements and SAP Best Practices/Processes. G. Conduct gap analysis of TO-BE process requirement and S/4HANA functionalities along with best practice solution to meet JPL NTPC requirements. H. The SI shall use the SAP S/4HANA training / mock environment to give training to the users to understand the working process of actual S/4HANA product. I. Business Solution Design document/ Blueprint: After detailed study, SI will prepare the solution design document / business blueprint (BBP) based on complete functional & technical requirements (including portal applications, Forms, all sorts of Reports, Interfaces, Customization, Enhancements etc.) and submit for JPL NTPC approval for the solutions which involve major modifications or new developments.

2.5.1.3	Realize Phase A. Development / Customization & Configuration: Based on the approved business blueprint, the SI will conduct the following activities. 1. Prepare and submit different codification details of master data in CVI etc. All Customer and Vendors in the existing ECC system shall be converted to Business Partners in S/4HANA. 2. Prepare functional & technical design documents, test scripts etc., and submit them for approval from JPL NTPC. 3. Configure and customize the S/4HANA as per approved business requirements. B. The SI shall modify and finalize the detailed project plan with a well-defined plan of action for all the activities as per the observations from the sandbox run. C. SI shall submit integration strategy document and develop necessary process for seamless integration between S/4HANA and other necessary applications of JPL NTPC to meet business requirements. D. Setup authorization matrix / user roles & responsibilities, and configuration in the system after approval from JPL NTPC. E. SI shall configure required roles during S/4HANA Implementation , in consultation with JPL NTPC. F. Activation of required SAP Business Functions after assessing the features and impacts in consultation with respective JPL NTPC functional core teams as per business requirement. G. Implementation of SAP notes as required by NTPC business processes. H. Carry out data migration activities including development of data migration templates, data cleansing, enrichment & transformation, data migration to configured solution. For this activity JPL NTPC will provide required access of existing ECC system to SI for extraction of required data which will be transported to S4 HANA system after required data cleansing, enrichment & transformation. I. Ensure all simplification items are managed, and adaptations are implemented. J. SI should ensure that all transactions and business processes of existing ECC should continue to work on the new platform of S/4HANA after implementation If any configuration/customization required same shall be carried out by the SI. K. Ensure SAP certified/approved data uploading tools are made available in line with existing LSMW/BDC in ECC. SI shall develop any new upload program required as per implemented solution. L. Develop/Activate & adopt identified Fiori apps in S/4HANA along with required customizations/validations/authorizations in consultation with JPL NTPC. Workflow integration

	is also to be done for FIORI apps as per JPL NTPC business requirement. Around 100 nos. of FIORI apps to be made available for JPL NTPC, which include transactional, analytical & custom apps and dashboards of different modules like PM, FI, MM, PS, HR etc. The list of FIORI apps to be developed/activated and workflow integration details shall be finalized in consultation with JPL NTPC. M. Developing/migrating all EP / Employee self-service (ESS) functionalities on Fiori UI as identified during Explore phase. N. Bar code or QR code scanning through mobile phone is to be implemented for tagging the technical objects in WCM module using FIORI apps. If any hardware or software is required for pilot implementation and testing, SI should supply them without any cost implication. O. If the desired 'TO- BE' process is not achievable through standard solution provided by SAP S/4HANA, SI will implement the same through enhancement or Z-development on business technology platform after discussion with JPL NTPC. P. All existing external systems should be integrated with the ERP solution on a consistent, on-line, real-time or batch processing basis in line with existing ECC integration and need to operate in an automatic manner without manual intervention. All required data residing in legacy applications shall be identified and transferred / integrated with S/4HANA solution. Q. Mock Runs: During implementation, SI shall conduct a minimum of Two mock runs of the newly installed system as per plan to ensure that the final run passes with zero bugs & exceptions and meet the downtime expectation. SI shall carryout/coordinate following activities during the process: 1. Ensure data integrity is fully maintained 2. Ensure the integration with other systems whichever is applicable and modules are working correctly as envisaged 3. High-level view of all major activities from ramp-down through ramp-up 4. Develop an end-to-end cutover procedure 5. Optimize the Cutover process downtime. 6. Validation of end-to-end cutover procedure 7. Ensure named and confirmed resources for each activity in the plan 8. Ensure the cutover and Go Live completes within the planned downtime. 9. Establish environments and process for testing 10. Technical and functional testing and issue resolution 11. Technical and functional support for testing 12. Prepare the documentation of all activities The above activities are to be achieved by the end of mock run cycles. R. Report of phase audit by OEM and compliance to be submitted by SI for closure of the phase.
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2.5.1.4	Deploy Phase A. Implementation of all existing data and new entries (if any) to S/4HANA system for all modules shall be carried out and confirmed by SI. All data is to be verified by JPL NTPC. The SI shall prepare and provide necessary Implementation check lists for verification by JPL NTPC. B. The process to be carried out by SI will include but not limited to the following: i) Identification of data uploading requirements ii) Collection and uploading of transaction data iii) Collection and uploading of master data iv) Closing or uploading of open transactions v) Collection and uploading of documents C. To achieve desired response time and process time, fine-tuning/optimization is to be done wherever required. D. Implementation and configuration of DMS . E. Entire S/4HANA and other solutions are to be tested and verified in by JPL NTPC in Sandbox, Development and Quality systems. Testing to be done based on pre-approved test scripts. The tests include Unit Tests, System Integration tests, performance/load test, user acceptance tests and security & access control tests. Test results are to be shared with JPL NTPC. The bidder shall document the results of the tests, fix the bugs/errors found during the testing, and submit a report to JPL NTPC. All pending or open issues should be closed before the end of this phase. F. User manuals are complete and approved. G. System configuration and other technical manuals are complete and approved. H. User support / help desk structure and process is established and ready to operate. I. Report of phase audit by OEM and compliance to be submitted by SI for closure of the phase. J. Finally, the SI shall present the project status to JPL NTPC management to take a go-live decision.
2.5.1.5	Go-Live / Production Phase During this phase, the SI will be required to carry out go-live activities after JPL NTPC clearance and as per the agreed approach and methodology. Before go-live, SI will ensure that following documents are provided to JPL NTPC (listis not exhaustive) 1. Configuration document consisting of system settings and parameters for each module (as applicable) 2. User manuals (module-wise) including system instructions with use cases, running of a program/T-code to perform specific task in the system with sample reports, screen formats etc. 3. Process flows and description manuals as necessary. 4. Standard Operation Procedure (SOP) manual. 5. Power user/ End user training details. 6. Data-migration as per cut-over plan, Data Migration results and sign-off documents. 7. Project Go-Live checklist report 8. System configuration and interface document 9. Help Desk support document. 10. Final preparation sign-off document.

	11. Pre-Go-Live issue log & resolution document 12. Any other document required for usage of implemented solution. 13. Phase audit by OEM and compliance report. On the successful completion of activities till Production environment, the SI commences Go-Live. SI should ensure vigilant monitoring on the performance of the system, growth of the database and system logs to detect the system level issues very early and provide quick resolution and preventive options. Upon successful operation of “SAP S/4HANA Solution” in the production environment, certificate will be issued by JPL NTPC for successful Go-Live. Go live shall be done within 9 months of award of contract for all the modules as per Scope of work. Post Go live stabilization support phase shall start after all the modules go live.
2.5.1.6	Three -month stabilization / Run Phase Post Go Live stabilization support will start from the date of Go live and continue for Three (3) months. During this period, the SI shall provide: 1. Post Go Live support period shall start from the date all the modules as envisaged in this PO goes Live. 2. Mandatory on-site support: In the stabilization period after Go-Live, SI shall continue to deploy the same Project Management, Technical and Functional staff who were involved in the implementation/migration phase, to ensure continuity of the key members. Any change in manpower composition shall be allowed only after ensuring the criteria mentioned in Sec 2.5.7 & 2.5.8 is meeting and written confirmation from JPL NTPC. The onsite consultants should have a sound knowledge of the deployed modules. 3. The SI shall provide functional support and maintenance of S/4HANA ERP Solution, during this period. The SI shall be responsible for rectifying all issues identified. 4. In the event of urgent requirements like upgrades, unresolved business showstoppers etc., the SI shall deploy additional consultants onsite without any additional charges. 5. During stabilization, if the solution is not working as per the requirements, the SI will rectify, reconfigure the SAP solution to comply with the same. 6. The SI need to update the required changes in user & configuration manuals/documentation and deliver to JPL NTPC prior to the completion of stabilization support. 7. If there is any change in the process or solution, knowledge transfer of the same to be given to the relevant functional and technical team of JPL NTPC. 8. SI shall ensure all agreed new processes are incorporated, which are not part of ECC. 9. SI shall implement any changes or new requirements relating to statutory compliances. 10. The SI shall provide support for ERP license management and control. 11. The stabilization period of three -months may be extended as required if any critical issues raised by JPL NTPC as part of implementation are left open. After the successful closure of all critical issues, JPL NTPC shall provide sign-off for the stabilization support phase.

2.5.2**Training Requirements**

Training shall be imparted to the JPL NTPC Team members during different phases of implementation as per requirement. During the Preparation phase, overview training to be provided to JPL NTPC key stakeholders and key users to familiarize them with features and functionalities of the new solution.

SI shall provide training to respective JPL NTPC core team members during implementation to create awareness of features of S/4HANA & and other solutions implemented. This will help JPL NTPC core team to understand the features of the SAP product and enable them to correlate with their respective functions while suggesting TO-BE processes. The training calendar for imparting training to Core teams, Power users, end users and top management shall be mutually finalized by both Project Managers during the Project Preparation Phase.

The following is an indicative training requirement for core team users of JPL NTPC based on their responsibilities.

Table-C

SN	Particular	Remarks
1	S/4HANA Functional training for Core Users (FICO, MM, SD, HR-HCM, Payroll, PM&WCM, QM, PS,)	25 users spread across all functions
2	HANA & ASE Administration Training and S/4HANA System Monitoring Training	2 users
3	Technical training HANA- ABAP & FIORI	2 users
4	Training on SAP BTP	2 users

SI shall conduct/coordinate training sessions, provide adequate training material and use Industry standard / OEM recommended training tools.

End-user training shall be conducted in multiple batches grouped according to functions.

Details of training requirements for different stakeholders of JPL NTPC are given in *Table D* below.

Table-D

SN	Particular	Remarks
1	Project Team/Core Team Training (off-line)	To be conducted by SI in module-wise batches for JPL NTPC functional and technical core teams as per details mentioned in <i>Table-C</i> of sec 2.5.2
2	Power User training for S/4HANA (online)	To be conducted by SI module-wise in multiple batches.
3	End Users Training for S/4 HANA	It shall be conducted by SI for all end users in multiple batches grouped according to functions.
4	Training for Senior Management (online)	Shall be conducted by SI in 2-3 batches covering different modules

	The meeting links for on-line power user training sessions shall be created by JPL NTPC in MS TEAMS or ZOOM and these meetings shall be recorded by JPL NTPC. In case link is provided by SI, the training session shall also be recorded by SI and shared with JPL NTPC. Exact training schedules and number of batches shall be finalized during the project preparation phase. JPL NTPC will measure the effectiveness of training imparted by SI after the completion of the training through training feedback forms. If the consolidated feedback on the training is not up to the desired level, the same shall be re-organized by the SI at no extra cost.
2.5.3	Project governance structure 1. SI will establish the Project Governance model during the Project Preparation phase in consultation with JPL NTPC. To facilitate communication between SI and JPL NTPC, a status review meeting shall be conducted regularly for planning and review of activities and resolution of open issues. 2. The intent of the governance structure is to provide: (a) visibility of project progress (b) interaction between the parties for resolving issues and decision-making (c) visibility of any elements of the project which may impact either party's ability to perform its duties. 3. The governance structure is intended to facilitate close working relationships between SI and JPL NTPC and may involve any third parties whose actions affect the ability to perform the Services. 4. Each party recognizes that the governance structures may evolve over time as the relationship matures and that any changes made to the governance structure shall be made in accordance with the change control procedure. 5. The relationship will be maintained at a peer-to-peer level between JPL NTPC and SI via scheduled and structured meetings and informal communication. 6. Day-to-day issues shall be resolved by regular interactions between relevant individuals within each work stream and work stream managers. Governance meetings shall focus on reviewing overall progress against objectives, managing escalated risks, and resolving escalated issues. 7. The decision made by a steering committee that would result in a change to this contract shall come into effect once the relevant change has been agreed via Change Request on mutually agreed terms and conditions. 8. All defined governance meetings shall have a pre-defined formal agenda and minutes must be recorded after each meeting. The SI shall prepare the agenda and organize the meetings. The agenda shall also cover issues based on the previous meetings and discussions with the customer. The SI shall prepare and share the minutes of the meeting. 9. The nominated representatives from JPL NTPC and SI may appoint or delegate a substitute who will have the authority to act on their behalf; both parties will take all reasonable steps to maintain continuity of their respective personnel within the governance structure. 10. JPL NTPC may appoint consultant who will be authorized to attend Steering committee meeting as well as progress review meetings in addition to JPL NTPC members. 11. Members of each governance body, and all attendees of governance meetings,

whether scheduled or ad hoc, should be of appropriate seniority and with appropriate levels of empowerment to take necessary decisions and actions.

12. Meetings of the various bodies within the governance structure may be held in person or by teleconference as agreed in advance between the parties. However, depending on the agenda the parties recognize that some meetings shall require physical presence and where this is required, the venue for the meetings shall be on the project's premises or at mutually agreed locations.
13. The SI will provide plans and status reports for OEM Services for discussion during the meetings.
14. It is suggested that SI and JPL NTPC shall implement the following meeting schedule to address project management and governance:

Table-E

Meeting	Frequency	SI Attendees	JPL NTPC Attendees
Progress Meeting	Weekly	Project Manager Solution Architect Team Leads as required	JPL NTPC Project Manager JPL NTPC Business Core Teams as required
Program Governance	Fortnightly	Program Manager Project Manager Solution Architect Team Leads	JPL NTPC Program Manager JPL NTPC Project Manager JPL NTPC JPL NTPC Business Lead (s) JPL NTPC ERP Lead (s)
Steering Committee	Monthly	Executive Sponsor Steering Committee Delivery Stakeholders Account Stakeholders	JPL NTPC Executive Sponsor JPL NTPC Business Stakeholders JPL NTPC ERP Stakeholders

The SI may modify the meeting model, structure and frequency in consultation with JPL NTPC during Preparation Phase.

2.5.4

Audit & Review by OEM

The SI will engage the Audit and Review services of OEM (i.e. SAP) for supporting the implementation/migration at his own cost and JPL NTPC shall not make any extra payment for the same. Audit and review by OEM must be done during Explore Phase and Go-Live phase of the Project (As per 2.5.6 Table "F")

2.5.5

Maintenance Support Services for five years after post go live stabilization period

	The SI shall provide Post Implementation Support services for the solution for 5 (five) years after the completion of post go live stabilization period. The support shall include following services: i. Supporting JPL NTPC ERP Core Teams through the following: a. Resolving end-user issues related to different modules b. Preparation of Functional Specifications/Technical Specifications c. Development/modification of processes d. Development/modification of reports e. Testing of developments/modifications f. Bug-fixing g. Preparation of user manuals etc. ii. Development/adoption of FIORI UI screens for various processes of different functions as per requirement. iii. Module Meetings/workshops (maximum of 12 sessions per year) by module-experts to apprise JPL NTPC Core Teams regarding new functionalities/ features/enhancements/developments in SAP ERP as per JPL NTPC’s requirement. iv. Expert consultancy support for any new SAP solution implementation, third party integration or enhancement of existing process in any module(s) as per JPL NTPC requirement 1. Key Personnel deployment during Post-Implementation Support phase Details provided in section 2.5.8 Table (H) & (I). 2. Change Management during Post-Implementation Phase The Change Management of JPL NTPC is a predefined process which keeps track of all the changes carried out in the system through proper documentation and approval mechanism.												
2.5.6	Project Deliverables (Bidder to develop Project Delivery Schedule beyond L2 Schedule as enclosed with specification). L3 , L4 Project Schedule should be reviewed & approved by JPL. <i>Table-F</i> <table><tr><th>No</th><th>Project Milestone</th><th>Deliverables (List not exhaustive)</th></tr><tr><td>1</td><td>Preparation Phase</td><td> 1. Project charter 2. Project Governance Structure 3. Resource mobilization plan 4. Training Plan 5. Exit Management Plan </td></tr><tr><td>2</td><td>Explore Phase</td><td> 1. Gap-Analysis Report 2. Business Blueprints of the new solution/major modifications 3. Training Report up to Explore Phase 4. OEM Audit and Compliance Report </td></tr><tr><td>3</td><td>Realize Phase</td><td> 1. User Acceptance Testing approach and test plan. 2. Completion certificate of Mock S4 Conversion run passes with zero bugs and exceptions. </td></tr></table>	No	Project Milestone	Deliverables (List not exhaustive)	1	Preparation Phase	1. Project charter 2. Project Governance Structure 3. Resource mobilization plan 4. Training Plan 5. Exit Management Plan	2	Explore Phase	1. Gap-Analysis Report 2. Business Blueprints of the new solution/major modifications 3. Training Report up to Explore Phase 4. OEM Audit and Compliance Report	3	Realize Phase	1. User Acceptance Testing approach and test plan. 2. Completion certificate of Mock S4 Conversion run passes with zero bugs and exceptions.
No	Project Milestone	Deliverables (List not exhaustive)											
1	Preparation Phase	1. Project charter 2. Project Governance Structure 3. Resource mobilization plan 4. Training Plan 5. Exit Management Plan											
2	Explore Phase	1. Gap-Analysis Report 2. Business Blueprints of the new solution/major modifications 3. Training Report up to Explore Phase 4. OEM Audit and Compliance Report											
3	Realize Phase	1. User Acceptance Testing approach and test plan. 2. Completion certificate of Mock S4 Conversion run passes with zero bugs and exceptions.											

			3. Completion certificate of Quality S4 Implementation n along with its integration with SAP and non- SAP application. 4. Completion certificate of UAT. 5. Integration testing Completion report. 6. Functional Specifications &Technical Specifications of new developments/modifications
3	Deploy Phase		1. Cut-over and Go-Live plan 2. Data Migration Strategy – a. Cut-Over strategy b. Data collection templates c. Data migration tools & conversions 3. Load and Stress Test Report 4. Completion certificate of all training needs as per Table-D of 2.5.2
8	Go live		1. Data migration completion report along with its integration with SAP and non- SAP applications. 2. OEM Audit Clearance for Go-Live readiness 3. Control reports before and after migration/Implementation 4. Configuration documents and technical documents for all process and modules. 5. Successful Go-Live Completion Report 6. Full System documentation including Configuration manuals, User manuals etc, and Knowledge transfer to JPL NTPC/ SAP Support Vendor wherever is applicable.
10	Three (3) Month Post Go-Live Stabilization (PGLS) Period		1. Publication of quarterly balance sheets from SAP for two quarters 2. Module wise sign-off including integration activities 3. Project Evaluation Report 4. Project Sign-off and Acceptance Report 5. OEM Review Report for Full Rollout and Stabilization
11	Five Years Support Service after Three months PGLS		1. Quarterly Activities Report in a mutually decided format 2. System health monitoring report. 3. Monthly Performance Report/MOM

2.5.7

Resource Plan

Proposed resource plan for different phases of this project is as follows:

Table-G

1	SAP S/4HANA Implementation	Onsite mode /Offsite mode
2	Support services during post go-live stabilization period	Onsite mode/ Offsite mode
3	Five years Maintenance Support Services for S4/HANA systems after post go-live stabilization period	Onsite mode /Offsite mode

	i) The SI shall ensure deployment of enough specialized and experienced manpower throughout the project to achieve the project objectives within time and with quality. ii) The SI needs to submit detailed CVs of each of the resource and Project Manager of the proposed project team. The Project team provided by the SI must be on their regular roll and SI shall certify the authenticity of their regular employment. The SI shall ensure to deploy the resources as per the details submitted in bid document with required experience and competency for each activity under the scope of this project. iii) On-boarding of the project team will be done on approval of CVs by JPL NTPC. Personal interaction of the project team members shall be done with JPL NTPC, if required for approval. iv) The SI shall provide contact numbers, email addresses, escalation matrix of the team. JPL NTPC shall be kept informed well in advance in case any changes are being made in the contact details. Any change of resources in the team should be submitted in writing at least one-month advance along with the alternative CVs (curriculum vitae) to the designated Project Manager of JPL NTPC for approval. Withdrawn resources shall be replaced with equally or more qualified and competent after approval from JPL NTPC. The SI shall be responsible for the knowledge transfer process and there shall not be any delay on the project delivery timelines on this account. v) JPL NTPC reserves the right to reject a non-performing resource in the team and on intimation from JPL NTPC, the SI shall arrange a replacement within 10 days without any additional cost to JPL NTPC. In such a case the SI shall provide a replacement, with a person of equivalent or better qualifications and experience. The detailed CV and profile of the resource should be submitted to JPL NTPC for approval. vi) The SI shall follow the JPL NTPC business calendar and working hours (6 -days work week is followed in JPL NTPC Plant or Noida Office). In case of any issues/emergency that hampers JPL NTPC business operations during holidays/non-working hours, adequate support mechanism should be in place to ensure the trouble-free business operations. vii) Expenses for travel, accommodation, phone charges, food and any other expenses for the resources will be borne by the SI. viii) The SI shall not subcontract, outsource, or engage any other third party for carrying out the work. ix) At no stage, the manpower deployment shall be less than that committed in the bid. Such manpower shall be maintained from start of the project up to complete Go-Live stage and further during Stabilization phase. x) The SI shall not make any changes to the composition of the Key Resources during the Term without prior written consent from JPL NTPC.												
2.5.8	Deployment of Resources A. Resource Qualification During Implementation Phase & Post Go-Live stabilization phase SI must propose a team consisting of experienced and skilled professionals with relevant experience in the proposed areas. The minimum qualification, experience and number of resources for deployment is given below: Table-H <table><tr><th>SN</th><th>Position</th><th>Minimum Qualifications and Experience Required for the position</th><th>Min Nos. of Resources (on-site)</th></tr><tr><td>1</td><td>Program Manager Unit Lead</td><td>Education: MBA/MCA/ M. Tech / B. Tech / B.E. / CA / ICWA or Equivalent</td><td>1</td></tr><tr><td></td><td></td><td>Total Experience: At least 15 years in SAP domain</td><td></td></tr></table>	SN	Position	Minimum Qualifications and Experience Required for the position	Min Nos. of Resources (on-site)	1	Program Manager Unit Lead	Education: MBA/MCA/ M. Tech / B. Tech / B.E. / CA / ICWA or Equivalent	1			Total Experience: At least 15 years in SAP domain	
SN	Position	Minimum Qualifications and Experience Required for the position	Min Nos. of Resources (on-site)										
1	Program Manager Unit Lead	Education: MBA/MCA/ M. Tech / B. Tech / B.E. / CA / ICWA or Equivalent	1										
		Total Experience: At least 15 years in SAP domain											

				Should have more than 7 years' experience as SAP projects Program Manager	
				Should have experience of at least two end-to-end SAP ECC or S/4 HANA projects as PMU Lead	
	2	Project Manager		Education: MBA/MCA/ M. Tech/ B. Tech / B.E. / CA / ICWA or equivalent	1
				Total Experience: At least 12 years in SAP domain	
				Should have more than 5 years' experience as SAP Project Manager	
				Should have experience of at least two end-to-end SAP ECC or S/4 HANA implementation/migration projects as Project Manager	
	3	Team Lead – HCM (HR)		Education: BE /B. Tech. /MCA /MBA/ Post Graduation in Human Resource Management or equivalent.	1
				Total Experience: 5 years in SAP Domain	
				Should have experience of more than 4 years in SAP-HCM (HR) in projects of similar nature with minimum two implementations.	
	4	Team Lead- Finance		Education: CA/ICWA/MBA or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-FICO in projects of similar nature with minimum two implementations.	
	5	Team Lead- SD		Education: CA/ICWA/MBA or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-SD in projects of similar nature with minimum two implementations.	
	6	Team Lead - Plant Maintenance		Education: BE /B. Tech. /MCA/MBA or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-PM in projects of similar nature with minimum two implementations.	
	7	Team Lead - WCM		Education: BE /B. Tech. /MCA/MBA or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-WCM in projects of similar nature with minimum two implementations.	

		8	Team Lead -MM	Education: BE /B. Tech. /MCA/MBA/CA/ICWA or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-MM in projects of similar nature with minimum two implementations.	
		9	Team Lead -Fuel Management (Custom)	Education: BE /B. Tech. /MCA/MBA/CA/ICWA or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-MM in projects of similar nature with minimum two implementations.	
		10	Team Lead Basis	Education: Full time MCA/ M. Tech/ B. Tech / B.E. or equivalent.	1
				Total Experience: At least 5 years in SAP domain	
				Should have experience of more than 4 years in SAP-BASIS in projects of similar nature with minimum two implementations.	
		11	Team Lead ABAP	Education: Full time MCA/ M. Tech/ B. Tech / B.E. or equivalent.	1
				Total Experience: At least 5 years in SAP domain	
				Should have experience of more than 4 years in SAP-ABAP in projects of similar nature with minimum two implementations.	
		12	Team Lead – Project System	Education: BE /B. Tech. /MCA /MBA/ Post Graduation in Human Resource Management or equivalent.	1
				Total Experience: 5 years in SAP Domain	
				Should have experience of more than 4 years in SAP-PS in projects of similar nature with minimum two implementations.	
		13	Team Lead- Quality Mgmt	Education: CA/ICWA/MBA/B.Tech/BE/M.Tech or equivalent.	1
				Total Experience: 5 years in SAP domain.	
				Should have experience of more than 4 years in SAP-QM in projects of similar nature with minimum two implementations.	
		14	Network and Security expert	Education: Full time MCA/ M. Tech/ B. Tech / B.E. or equivalent.	1 (Shared)
				Experience of more than 4 years in Network & Security domain.	

15	Functional Consultants- for the following modules: FICO, MM, HCM& Payroll, SD, PM, WCM, , PS, , QM, TRM , FM	Education: CA / ICWA / MBA / MCA/ M. Tech/ B. Tech / B.E. or equivalent in relevant area.	As per requirement to meet project objectives and timelines.
		Standard modules: Experience of more than 3 years as consultant in respective modules of SAP.	
16	Technical consultants in ABAP, FIORI, BASIS, DMS	Custom modules: for FM module, consultant should have experience of more than 3 years in MM module whereas for Operation module, consultant should have experience of more than 3 years in QM module.	ABAP - 2 FIORI - 1 BASIS- 1 DMS- 1
		Education: Full time MCA/ M. Tech/ B. Tech / B.E. or equivalent.	
		Experience of more than 3 years as consultant in respective modules.	

Note: The resource requirement given in the above table is indicative. However, SI shall deploy additional manpower based on requirement to meet project objectives and timeline without any additional financial implication to JPL NTPC.

B. Deployment of Resources During Post-Implementation Support Phase (five years)

Resource deployment during the five-year support period shall include both on-site and off-site mode of deployment. SI shall deploy the Project Manager (Project In-charge), Technical & Functional Experts and Other Support Staff during the 5-year support Phase after PGLS. The deployed resources shall work from Monday to Saturday as per JPL NTPC business hours. Each on-site deployed resource is entitled to 18 days of leave per year. In case the deployed resource is going on leave for more than 10 business days, SI shall provide an alternate resource, with approval from JPL NTPC. More details on resource deployment during the five-year support phase are mentioned in section 10.2. The minimum resource deployment requirement post implementation support phase (five years) is given in table below:

Table-I

SI No	Position	Minimum Qualifications and Experience Required for the position	Min No. of Resources Required
1	Project Manager	Education: MBA/MCA/ M. Tech / B. Tech / B.E. / CA / ICWA or equivalent	1 no. (Off-site)

			Total Experience: At least 10 years in SAP domain	
			Should have more than 4 years of experience of handling SAP Project Manager.	
	2	Consultants for FICO, MM, HCM, SD	Education: CA / ICWA / MBA / MCA/ M. Tech/ B. Tech / B.E. or equivalent in relevant area.	Total 4 nos. of 1 st Year <i>On-site subsequently On-site/Off site as per requirement</i> Resources: 1 no. per module
			Experience of more than 3 years as consultant in respective modules of SAP.	
	3	Consultants for , PM&WCM, QM, PS, , TRM , FM	Education: CA / ICWA / MBA / MCA/ M. Tech/ B. Tech / B.E. or equivalent in relevant area.	Shared resources Onsite/Off-site resources (0.5 no. per module)
			Experience of more than 3 years as consultant in respective modules of SAP.	
	3	ABAP & Fiori Consultants	Education: MCA/ M. Tech/ B. Tech / B.E. or equivalent in relevant area.	2 nos. (<i>On site</i>) for first year
			Experience of more than 3 years as consultant in respective area of SAP.	2 nos. (<i>Off site</i>) from second year onwards
	Resource for all other modules which is not specifically mentioned above shall be made available as per requirement			
	Note: One dedicated resource onsite (Plant) shall always be available to coordinate between JPL NTPC Core team and support team of SI.			
3.0	Network & Security			
3.1	Scope of Work related to Network and Security The SI shall implement the additional Network and Security elements in the JPL NTPC RISE hyperscaler environment as per details in Annexure-1. The list of required services given in Annexure-1 are in the scope of SI.			
4.0	Other Conditions			
	The SI's responsibility shall also include the following i) SI shall ensure that during implementation, the performance, security, etc. of the existing setup is not compromised. ii) All software, testing tools, etc., as required during all operations such as design, development, installation, conversion, testing, roll-out etc., shall be provided by the SI at no extra cost to the owner for completing the scope of work as per this specification. iii) All the software, tools etc. used during implementation/conversion should be genuine and should have a valid license and OEM recommended/certified.			

	iv) The bidder shall ensure that none of JPL NTPC equipment/structure/setup gets damaged during commissioning and operations of the supplied equipment. Any damages, if caused to JPL NTPC property due to Bidder's activity/negligence or the operations of the equipment, shall be rectified, repaired or replaced by the bidder at their own cost. v) The Bidder shall depute an experienced onsite Project Manager during implementation. The onsite Project Manager will be totally responsible for the completion of the project till handover to owner and will report to JPL NTPC Engineer-In-Charge. vi) The SI shall provide contact numbers, email addresses, escalation matrix of the support team. JPL NTPC shall be kept informed well in advance in case of changes in contact details of SI resources. JPL NTPC shall be notified for any change of deployed resources in the support team in writing at least one-month advance. Proper Knowledge Transfer (KT) should be done which shall be verified by JPL NTPC. vii) The SI shall follow the JPL NTPC business calendar and working hours. (Applicable to both offsite and onsite resources). In case of any issues/emergency that hampers JPL NTPC operations during holidays/non-working hours; adequate support mechanisms shall be in place to ensure the trouble-free business operations. viii) The SI shall, at his own cost, arrange for necessary insurance to cover all the risks assumed by the SI under this contract in respect of its deputed personnel for the execution of this contract during the entire period of the contract. JPL NTPC shall have no liability against any loss/ damage/ injury etc. to its personnel or any other concerned regarding whatsoever. ix) The SI shall not subcontract, outsource, or engage any other third party. x) Expenses for travel, phone charges, food, and accommodation etc. for the resources will not be borne by JPL NTPC. xi) Safety Compliance for Plant: For SI Personnel deputed at Plant Sit 1. SI must provide all PPEs (Safety shoes, Helmet, safety Belt, Mask etc). Strong action will be taken for violation of industrial safety guidelines as per rules. 2. For gate pass , all worker will go Safety training and medical check-up as per NTPC JPL/ and need to follow state/central Govt. laws 3. Insurance of deployed manpower shall be in scope of the agency as per rules. 4. Accommodation of deployed manpower in this contract is not in the scope of NTPC-JPL. Vendor has to ensure the accommodation.
5.0	SITE PREPARATION
5.1	Facilities to be provided by JPL NTPC: Limited space along with LAN/internet connection at JPL NTPC premises for SI's personnel during implementation and support period. Laptop and other computing facilities for the deployed personnel shall be provided by SI. The SI is requested to visit JPL NTPC Plant and Noida Office to understand the logistics requirement before bid submission.
6.0	BID EVALUATION The technical part of the bid shall be opened first which should also include QR and QCBS formats dully filled along with required documents. The commercial bid shall be opened of the bidders whose technical and mandatory QR criterion are found acceptable and shall be further evaluated for total package based on QCBS method as mentioned in bid evaluation criterion attached as Annexure – 3
7.0	Payment Terms

	The cost of ERP implementation services including training, data migration, data entry etc., for the ERP Solution will be paid based on the completion of the key milestones of the project as identified and acceptance of the deliverables associated with the milestones by JPL NTPC. The cost of support shall be paid at the end of each quarter after due verification of deliverables and quarterly activities report by JPL NTPC. The payment terms for S/4HANA implementation services and post implementation five-year support services are mentioned in section 7.1 & 7.2 respectively.																		
7.1	Contract 1 – Payment Terms for S/4HANA Implementation Services (9 months implementation + 3 month post go Live period). No advance payment is envisaged. <i>Table-J</i> <table><tr><th>Milestone</th><th>Description</th><th>Payment</th></tr><tr><td>Milestone1</td><td>Completion of Explore Phase</td><td>10% of the cost of implementation Service</td></tr><tr><td>Milestone2</td><td>Completion of Realize Phase</td><td>20% of the cost of implementation Service</td></tr><tr><td>Milestone3</td><td>Completion of Deploy Phase</td><td>20% of the cost of implementation Service</td></tr><tr><td>Milestone4</td><td>Go-Live Phase</td><td>30 % of the cost of implementation Service</td></tr><tr><td>Milestone5</td><td>Completion of post Go-live stabilization phase</td><td>20% of the cost of implementation Service</td></tr></table>	Milestone	Description	Payment	Milestone1	Completion of Explore Phase	10% of the cost of implementation Service	Milestone2	Completion of Realize Phase	20% of the cost of implementation Service	Milestone3	Completion of Deploy Phase	20% of the cost of implementation Service	Milestone4	Go-Live Phase	30 % of the cost of implementation Service	Milestone5	Completion of post Go-live stabilization phase	20% of the cost of implementation Service
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Milestone5	Completion of post Go-live stabilization phase	20% of the cost of implementation Service																	
7.2	Contract 2 – After Post Go Live (After Completion of Contract 1) Payment Terms for S/4HANA Maintenance Support Service (5Years) <i>Table-K</i> <table><tr><th>S N</th><th>Activity</th><th>Payment Terms</th><th>Payment Milestones</th></tr><tr><td>1</td><td>Maintenance Support Services</td><td>Quarterly payments at the end of each quarter 95% of Quarterly Billing with 100% Taxes within 30 days. Retention : 5% will be released on completion on yearly basis</td><td>On certification of EIC</td></tr></table>	S N	Activity	Payment Terms	Payment Milestones	1	Maintenance Support Services	Quarterly payments at the end of each quarter 95% of Quarterly Billing with 100% Taxes within 30 days. Retention : 5% will be released on completion on yearly basis	On certification of EIC										
S N	Activity	Payment Terms	Payment Milestones																
1	Maintenance Support Services	Quarterly payments at the end of each quarter 95% of Quarterly Billing with 100% Taxes within 30 days. Retention : 5% will be released on completion on yearly basis	On certification of EIC																

	Note :- As mentioned in section 1.3(iv) the quoted price for five-year support should not be less than 20% of the total bid price. If it is less than 20% then the difference between actual amount and 20% of bid price shall be withheld and released proportionately at the end of each year of support.																													
7.3	Contract 3 - Payment Terms for supply of S/4HANA on RISE Table-L <table><tr><th>S N</th><th>Activity</th><th>Duration</th><th>Payment Terms</th><th>Payment Milestones</th></tr><tr><td>1</td><td>supply of S/4HANA on RISE</td><td>5-Years</td><td>Quarterly payments at the end of each quarter</td><td>On certification of EIC</td></tr><tr><td></td><td>Project Implementation phase 1-Year (12 months=9+3)</td><td></td><td>95% Invoice Value with 100% taxes Within 30 days from date of Mobilization of Team at Site.</td><td></td></tr><tr><td></td><td>AND</td><td></td><td></td><td></td></tr><tr><td></td><td>Maintenance Support phase (4-Years)</td><td></td><td>Retention: 5% will be released on yearly basis</td><td></td></tr></table> Note :- As mentioned above, section 1.3(v) the quoted price for supply of S/4HANA on RISE for five-year should not be less than 40% of the total bid price. If it is less than 40% then the difference between actual amount and 40% of bid price shall be withheld and released proportionately at the end of each year.					S N	Activity	Duration	Payment Terms	Payment Milestones	1	supply of S/4HANA on RISE	5-Years	Quarterly payments at the end of each quarter	On certification of EIC		Project Implementation phase 1-Year (12 months=9+3)		95% Invoice Value with 100% taxes Within 30 days from date of Mobilization of Team at Site.			AND					Maintenance Support phase (4-Years)		Retention: 5% will be released on yearly basis	
S N	Activity	Duration	Payment Terms	Payment Milestones																										
1	supply of S/4HANA on RISE	5-Years	Quarterly payments at the end of each quarter	On certification of EIC																										
	Project Implementation phase 1-Year (12 months=9+3)		95% Invoice Value with 100% taxes Within 30 days from date of Mobilization of Team at Site.																											
	AND																													
	Maintenance Support phase (4-Years)		Retention: 5% will be released on yearly basis																											
7.4	Contract 4 - Payment Terms for Lift and Shift of existing SAP ECC system and Supply and Maintenance charges of Cloud Infra setup for hosting it for next 5 Year in a limited access mode <table><tr><th>S N</th><th>Activity</th><th>Payment Terms</th><th>Payment Milestones</th></tr><tr><td>1</td><td>Lift and Shift of existing SAP ECC system</td><td>100 %</td><td>On certification of EIC</td></tr><tr><td>1</td><td>Supply and Maintenance charges of Cloud Infra setup for hosting it for 5</td><td>Quarterly payments at the end of each quarter</td><td>On certification of EIC</td></tr></table>					S N	Activity	Payment Terms	Payment Milestones	1	Lift and Shift of existing SAP ECC system	100 %	On certification of EIC	1	Supply and Maintenance charges of Cloud Infra setup for hosting it for 5	Quarterly payments at the end of each quarter	On certification of EIC													
S N	Activity	Payment Terms	Payment Milestones																											
1	Lift and Shift of existing SAP ECC system	100 %	On certification of EIC																											
1	Supply and Maintenance charges of Cloud Infra setup for hosting it for 5	Quarterly payments at the end of each quarter	On certification of EIC																											

		Year in a limited access mode		
8.0	Contract Period i) Contractual delivery period for Implementation of S/4 HANA, Payroll and other components as per SoW, will be 9 months from the date of LOA. ii) The Post Go-Live Support during Stabilization period will be for a period of 3 months from the date of actual Go-Live. iii) Post implementation support shall be for 5 years (60 months) from the actual end date of Post Go-Live Stabilization period. iv) The overall duration of the contract shall be for a period of 6 Years (72 Months) from the date of issuance of LOA.			
9.0	Knowledge Transfer (KT) Procedure Knowledge transfer shall be carried out to the new resource / NTPC team in all respect on completion of the contract period. The duration of the KT will be mutually decided by JPL NTPC and SI.			
10.0	Penalties for non-compliances			
10.1	Penalties during implementation and post Go-Live stabilization phase i) If Project Manager/any Team Lead is not deployed or do not have adequate qualification/experience as per the requirement, a deduction of Rs 15,000/- per day shall be made for the entire duration till deployment of suitable manpower. ii) If any resource is found to be non-performing, JPL NTPC reserves the right to ask SI for a replacement. SI shall provide a suitable replacement within 10 business days. If suitable replacement is not provided within the period by SI, a deduction of Rs 10,000/- per day shall be made for the entire duration from the date of JPL NTPC request for replacement. iii) Further, in case of non-deployment of resources or deployment of resources with inadequate qualifications/ experience, JPL NTPC reserves the right to engage suitable resources, including from OEM, at the cost of the SI.			
10.2	Penalties during five-year support period There is a mixed mode of deployment in the five-year support period, which includes both on-site and off-site resources. A. Conditions for <i>off-site</i> resources: The <i>dedicated off-site</i> resources are expected to be available on call during JPL NTPC office hours. In case of any issues/emergency that hampers JPLNTPC business operations during holidays/non-working hours, adequate support mechanism should be in place to ensure the trouble-free business operations. In case of non-availability of assigned dedicated resources, prior information to be provided to JPL NTPC along with the contact number of an alternative person. If SI fails to provide alternative person during absence of <i>off-site</i> consultant, a penalty of Rs.5000/- shall be deducted per day. B. Conditions for <i>on-site</i> resources			

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	4	C4	Lift and Shift of Existing SAP ECC system to a suitable cloud environment in a limited access mode	Lum-Sum	1
	5	C4	Supply and Maintenance charges of Cloud Infra setup for hosting existing SAP ECC system for 5 Year after GO-Live of new system	Quarter	20
12.0	Annexures				
	Annexure- 1		Security and Compliance Audit Scope (2.3.2 Point: 34) for SAP S/4 HANA Private Cloud		
	Annexure- 2		FRS (Functional requirement specification)		
	Annexure- Bid Evaluation Criteria		Bid evaluation Criterion (Refer 1NITR2 page 7 till Page 17)		
	Annexure-4		Backup and retention policy of JPL NTPC ERP Server		
	Annexure-5		Minimum SAP license / configuration (Indicative)		

*** - Annexure 3 – is not attached / available.